Optimization in the Länd

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Book of Abstracts

25–27 November 2026

University of Mannheim, Germany

Joint meeting of the IFIP TC7 working groups
WG 7.4 (Inverse Problems) and WG 7.8 (Nonlinear Optimisation)

Organizers

Mathias Staudigl · Roland Herzog · Simon Weißmann
Andrea Ebner · Meggie Marschner · Siqi Qu · Johannes Schnebel

Venue

Wednesday: Room 008.1, building B6 30–32, 68159 Mannheim
Thursday & Friday: Fuchs-Festsaal (O 138), Mannheim Palace (Schloss)

https://jschnebel.github.io/optimization-laend.de/

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Wednesday, 25 November · 13:00 – 13:45

Adjoint-based Methods for Optimal Control of Hyperbolic Systems of Conservation Laws

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Keywords: optimal control, adjoint equation, hyperbolic systems of conservation laws

Hyperbolic systems of conservation problems arise in many applications, for example in fluid mechanics or traffic flow models. A particular feature lies in the fact that even for smooth data discontinuities may develop (shocks, contact discontinuities). This complicates the development of derivative based optimality conditions and solution methods in the context of optimal control and inverse problems governed by these PDEs.

We consider strictly hyperbolic $n \times n$ systems of conservation laws in one space dimension and present a sensitivity calculus for the optimal control of piecewise C^1 solutions containing shocks. This implies the differentiability of many objective functionals with respect to controls. To obtain an adjoint-based derivative representation of the objective functional, we study the adjoint equation, which is a system of transport equations with discontinuous coefficients. It turns out that the solution of the adjoint equation is non-unique and an appropriate boundary condition along shocks is required to select the correct adjoint state. We present existence, uniqueness and regularity results for the adjoint equation.

Finally, we discuss challenges and possible approaches to solve the adjoint equation numerically. Moreover, we derive from the properties of the adjoint state conclusions for the design of structure-exploiting derivative based methods for optimal control and inverse problems involving systems of conservation laws.

Acknowledgements. This work was supported by the DFG in the "Collaborative Research Centre CRC/Transregio 154, Mathematical Modelling, Simulation and Optimization Using the Example of Gas Networks", Project-ID 239904186, Projects A02.

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Wednesday, 25 November · 13:45 – 14:10

Learning Optimization Algorithms with Average Case Convergence Rates

Peter Ochs

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The change of paradigm from purely model driven to data driven (learning based) approaches has tremendously altered the picture in many applications in Machine Learning, Computer Vision, Signal Processing, Inverse Problems, Statistics and so on. There is no need to mention the significant boost in performance for many specific applications, thanks to the advent of large scale Deep Learning. In this talk, we open the area of optimization algorithms for this data driven paradigm, for which theoretical guarantees are indispensable. The expectations about an optimization algorithm are clearly beyond empirical evidence, as there may be a whole processing pipeline depending on a reliable output of the optimization algorithm, and application domains of algorithms can vary significantly. While there is already a vast literature on "learning to optimize", there is no theoretical guarantees associated with these algorithms that meet these expectations from an optimization point of view. We develop the first framework to learn optimization algorithms with provable generalization guarantees to certain classes of optimization problems, while the learning based backbone enables the algorithms' functioning far beyond the limitations of classical (deterministic) worst case bounds. Our results rely on PAC-Bayes bounds for general, unbounded loss-functions based on exponential families. We learn optimization algorithms with provable generalization guarantees (PAC-bounds) and explicit trade-off between a high probability of convergence and a high convergence speed.

Wednesday, 25 November · 14:10 – 14:35

Estimation of Monge Transport Maps with Empirical Brenier Potentials

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Keywords: Optimal Transport, Statistical Estimation, Monge Problem, Error Bound

We analyze a statistical estimator for Monge transport maps: solutions to the quadratic optimal transport problem. For absolutely continuous source measures, this map is uniquely defined as the gradient of a convex function, a result known as Brenier theorem. Without absolute continuity, the problem is relaxed, maps are replaced by coupling measures, and optimal couplings are supported on the subdifferential of a convex function: a Brenier potential. This characterization is the basis for our Monge transport map estimator, for measures known only through finite samples. The resulting Brenier potential has a simple closed form expression based on the dual solution of the discrete sampled problem. We exhibit convergence rates for this estimator based on a new error bound for the dual quadratic optimal transport problem. Our methodology does not rely on smoothness or continuity of the Monge transport map and requires no computation beyond primal-dual solutions of the discrete, finite dimensional, linear program transport problem.

Wednesday, 25 November · 15:05 – 15:30

Zeroth Order Proximal Point Algorithm

Silvia Villa

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Abstract: TBA

Wednesday, 25 November · 15:30 – 15:55

Semismooth Newton Methods with Global Convergence Rates

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Keywords: optimization, second-order methods, semismoothness, convergence rates

We propose LeAP-SSN (Levenberg–Marquardt Adaptive Proximal Semismooth Newton method), a semismooth Newton-type method with a parameter-free globalisation strategy that guarantees, for minimization problems, nonasymptotic sublinear convergence rate in convex setting and linear convergence rate under the Polyak–Łojasiewicz condition, in Hilbert spaces. The method employs an adaptive Levenberg–Marquardt regularisation for the Newton steps, combined with backtracking, and does not require knowledge of problem-specific constants. The algorithm achieves superlinear convergence under mild semismoothness and Dennis–Moré or partial smoothness conditions. By combining strong global guarantees with superlinear local rates in a fully parameter-agnostic framework, LeAP-SSN bridges the gap between globally convergent second-order algorithms and Semismooth Newton's methods with the fast asymptotic local rates. The practical efficiency of the method is illustrated on representative problems from imaging, contact mechanics, and machine learning. We also propose a modification of this method based on gradient regularization and show that it achieves improved local superlinear convergence under strong semismoothness and is able to work with lazy Hessian updates. The details can be found in https://arxiv.org/abs/2508.16468 and https://arxiv.org/abs/2602.08069.

Acknowledgements. This work was supported by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy – The Berlin Mathematics Research Center MATH+ (EXC-2046/1, project ID: 390685689).

Wednesday, 25 November · 15:55 – 16:40

Nonlinear Preconditioning for Optimization and Monotone Inclusions

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Keywords: optimization, machine learning

Many algorithms used in modern optimization, such as gradient clipping, sign-based and spectral optimizers like Muon, as well as adaptive schemes for (L_0, L_1)-smooth problems, can be understood through the lens of nonlinear preconditioning in the dual space. Here, the search direction is determined by passing the gradient, or more generally the operator, through the gradient of a Legendre convex reference function. In this talk I will give an overview of my group's recent work on this idea. I will start with minimization, where the theory is now well developed, thanks to tools of generalized convexity and duality [1, 2, 3, 4, 5, 6]. I will then move to monotone inclusion problems, where many basic questions are still open [7, 8].

Acknowledgements. This work was supported by the Research Foundation Flanders (FWO) PhD grant 11A8T26N and research projects G081222N, G033822N, and G0A0920N; Research Council KUL grant C14/24/103.

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Wednesday, 25 November · 17:00 – 17:45

Calmness of the Solution-Set Mapping for Linear Bilevel and Pricing Problems

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Keywords: bilevel optimization, linear bilevel optimization, pricing problems, approximate feasibility, calmness, outer semicontinuity

We study linear bilevel and pricing problems in which the upper- and lower-level constraints' right-hand sides are perturbed. In this setting, it is an important question, also for the validity of numerical solution schemes, if the solution-set mapping of the parametric bilevel problem is calm at the zero-perturbation. We provide the complete picture both for linear bilevel as well as for pricing problems. If the result is positive or not depends on whether the problems have coupling constraints or not and on whether the perturbation is allowed to take place in both levels and if they lead to relaxations or tightenings of the respective constraints. In particular, the solution-set mapping is calm for linear bilevel problems without coupling constraints and for pricing problems if the upper-level problem is not tightened by the perturbation. For the negative results, we provide illustrative counterexamples.

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Wednesday, 25 November · 17:45 – 18:10

An Invertible Generative Model for Forward and Inverse Problems

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Keywords: Bayesian inverse problems, normalizing flows, triangular transport, amortized inference

Many inverse problems require both a forward simulator, which generates data for given parameters, and an inference procedure, which characterizes the parameters compatible with observed data. In a Bayesian formulation these tasks amount to sampling from the likelihood and the posterior, respectively. Both tasks can benefit from recent advances in generative models. Existing amortized approaches typically learn separate conditional generative models for the two directions [3], or focus primarily on the inverse direction [1].

We introduce a single invertible generative map that performs both tasks. Starting from the classical use of triangular transport maps for conditional sampling [2], we combine a lower-triangular transport, adapted to the likelihood, with an upper-triangular transport, adapted to the posterior [4]. The resulting bijection acts on the joint parameter–data space: evaluated forward, it generates samples from the likelihood for a prescribed parameter; evaluated in reverse, it generates posterior samples for prescribed data. We investigate the (non-)uniqueness of this construction, and propose a variational formulation that allows us to learn the map from samples.

In linear Gaussian problems the construction is explicit and, unlike the individual triangular transports, can remain well conditioned as the observational noise vanishes. Nonlinear examples and an image-inpainting experiment illustrate simulation and uncertainty-aware inversion with the same model. The framework thus offers a unified route to amortized forward simulation and Bayesian inference, with potential extensions to experimental design.

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Thursday, 26 November · 08:35 – 09:00

TBA

Jalal Fadili

ENSICAEN, Normandie Université, Caen

Thursday, 26 November · 09:00 – 09:45

On the Interplay Between Continuous- and Discrete-Time Methods in Convex Optimization and Monotone Inclusions

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Keywords: continuous-time methods, algorithms, convex optimization, monotone inclusions, fast convergence rates

In this talk, we explore the interplay between continuous-time and discrete-time models for solving convex optimization problems and monotone inclusions. We discuss strategies for achieving faster convergence rates across the different settings, as well as issues related to the convergence of trajectories and iterates.

Thursday, 26 November · 09:45 – 10:10

Inexact Interior-Point Proximal Methods for Cone-Constrained Nonsmooth Optimization

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Keywords: proximal methods, interior-point methods, PDE constrained optimization, complexity

We present an inexact interior-point method for a class of nonsmooth nonconvex optimization problems with conic inequality constraints. The problem is posed in a Hilbert space, while the inequality constraint is induced by an order cone in a Banach lattice. The objective combines a smooth term with a convex nonsmooth term whose proximal mapping is computable. The method is based on barrier-regularized subproblems, which are solved inexactly by a proximal-gradient scheme.

We consider both logarithmic and power-type barrier functions and establish the analytical properties required for the convergence and complexity analysis. Under suitable constraint qualifications, we show that the primal iterates and the associated barrier multipliers satisfy quantitative approximate KKT conditions. Based on these estimates, we establish convergence to stationary points and derive complexity bounds for the full outer–inner scheme, combining the barrier-parameter updates with the inexact proximal-gradient solution of the subproblems. Numerical examples from finite-dimensional nonlinear optimization and PDE-constrained optimization with pointwise state constraints illustrate the theoretical results.

Thursday, 26 November · 10:35 – 11:20

How to Make (Bregman-)Kaczmarz Methods Fast

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Keywords: constrained optimization, Kaczmarz methods

Bregman-Kaczmarz methods can compute solutions of systems of linear equations that additionally minimize a strongly convex functions (e.g. minimum-norm solutions, but also sparse solutions). Their plain version is oftentimes quite slow and hence, there is interest in speeding up the methods. Optimal step-sizes are, on the one hand, often hard to compute and, on the other hand, do not always bring much improvements for. In this talk I will treat several acceleration mechanisms e.g. based on momentum (with a computable approximate minimal error criterion) and dual acceleration [1, 2, 3].

Acknowledgements. This work was supported by the International Training Network TraDE-OPT funded by MSCA, grant agreement ID: 861137.

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Thursday, 26 November · 11:20 – 11:45

Point Source Localisation with Unbalanced Optimal Transport

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Keywords: optimization, inverse problems, measure spaces, convection–diffusion

Replacing the quadratic proximal penalty familiar from Hilbert spaces by an unbalanced optimal transport distance, we develop forward-backward type optimisation methods in spaces of Radon measures. We avoid the actual computation of the optimal transport distances through the use of transport three-plans and the rough concept of transport subdifferentials. The resulting algorithm has a step similar to the sliding heuristics previously introduced for conditional gradient methods, however, now non-heuristically derived from the geometry of the space [2].

Afterwards, as an application of the above algorithm, we study the inverse problem of locating gas leaks from line-of-sight concentration measurements using a convection–diffusion model with the source term a Radon measure. By imposing sparsity-promoting regularisation on this measure, we recover point sources—identifying both their locations and intensities—rather than diffuse approximations. We jointly estimate the underlying physical convection (wind) and diffusion parameters. Our main theoretical contribution is the stability analysis of the convection–diffusion equation with respect to its parameters: the measure, and the convection and diffusion fields. Our experiments demonstrate accurate localisation, highlighting the potential of the method for practical gas emission detection [1].

Acknowledgements. This work was supported by the Jane and Aatos Erkko Foundation.

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Thursday, 26 November · 11:45 – 12:30

Infinite Infimal Convolution Regularization: Concepts, Models and Optimization

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Keywords: optimization, inverse problems, regularization functionals, infimal convolution

Variational methods are a well-established, powerful framework for solving inverse problems. Typically, they combine a discrepancy term that models the forward problem and noise with a regularization term that incorporates prior knowledge. A wide range of regularizers has been proposed in the literature, from classical Tikhonov penalties to modern data-driven and learned functionals. In this context, the infimal convolution of regularization functionals is a well-established tool to balance finitely many such penalties.

In this talk, we introduce infinite infimal convolution as a way to select from a continuum of regularization penalties. This leads to a flexible, data-adaptive model selection mechanism based on an infinite, parameterized family of regularizers. We formulate infinite infimal convolution regularization as a variational problem over Banach-space valued Radon measures and analyze well-posedness and convergence in the setting of linear inverse problems. The framework covers various modeling scenarios, including regularization across smoothness scales and parameterized families of functionals [1]. We further illustrate its use in less direct applications, such as relaxed bilinear inverse problems and the learning of certain classes of regularization functionals. On the optimization side, we show how inverse problems with infinite infimal convolution regularization can be solved using generalized conditional gradient methods [1, 2]. Finally, we explain how the arising subproblems can be treated efficiently via extremal point techniques, yielding practical algorithms with convergence guarantees.

This is joint work with Marcello Carioni, Silvio Fanzon, Martin Holler, Yury Korolev, Carola-Bibiane Schönlieb, and Daniel Walter.

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Thursday, 26 November · 13:30 – 14:15

Lifting Input-Convex Neural Network Regularizers

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Keywords: variational reconstruction, input-convex neural networks, primal-dual algorithm, convex reformulation, image restoration

We consider inverse problems $y = Ax + \varepsilon$, where y represents the observed data, A is a known forward operator, and ε is additive noise. Regularization can be approached through variational methods, i.e., we aim to solve

$$\min_{x \in X} \{D(Ax, y) + R_\theta(x)\},$$

where D represents a convex data-fidelity function and R_θ is an Input-Convex Neural Network (ICNN) [1, 4]. An L-layer ICNN parameterizes the regularizer $R_\theta : X \to \mathbb{R}$ recursively via hidden layer states z^ℓ: starting from $z^1 = \sigma(W^0 x + b^0)$, we define

$$z^{\ell+1} = \sigma\left(V^\ell z^\ell + W^\ell x + b^\ell\right), \qquad \ell = 1, \dots, L-2,$$

and then the regularizer can be evaluated as

$$R_\theta(x) = \sigma\left(V^{L-1} z^{L-1} + W^{L-1} x + b^{L-1}\right).$$

To guarantee that R_θ is globally convex with respect to its input x, the network parameters are constrained such that the feed-forward matrices V^ℓ contain only non-negative entries, and the activation function σ is convex and non-decreasing. However, the nested compositions may lead to vanishing subgradients and difficult-to-compute proximal mappings, which motivates the present work.

The core contribution of this work lies in decoupling the structural constraints of the deep network architecture. Instead of treating the objective implicitly through composition, we lift the optimization space by defining auxiliary variables corresponding to the network layers. By exploiting the mathematical properties of the epigraph of the layer-wise convex activation functions, we show that the original nested problem is mathematically equivalent to a decoupled optimization problem over the original unknown x and the full set of multi-layer activations $z = (z^1, \dots, z^{L-2})$. Because this formulation eliminates the nested composition, the objective separates into isolated proximal-friendly terms and linear blocks. The optimization can be compactly cast into the standard saddle-point format and solved with standard algorithms such as the Primal-Dual Hybrid Gradient (PDHG) algorithm [2]. Numerical experiments on image restoration tasks demonstrate that the proposed method has advantages in terms of speed and reliability over existing approaches which consider nested compositions.

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Thursday, 26 November · 14:15 – 14:40

Adaptive Block-Coordinate Forward-Backward: Particular Case of Multilevel Block Strategy for Image Reconstruction

Nelly Pustelnik

ENS de Lyon

Classical first-order optimization methods for imaging inverse problems struggle to scale efficiently with increasing image resolution. Here, we introduce an adaptive multiresolution block coordinate Forward-Backward algorithm for image restoration. The selection of multiresolution blocks is guided by the local magnitude of the proximal update, using a stochastic, non-smooth Gauss-Southwell rule applied to the image wavelet decomposition. This adaptive approach dynamically balances updates across scales, prioritizing coarse or fine blocks based on the degradation characteristics. As a result, the proposed method automatically adjusts to varying levels of blur and noise without requiring a predefined hierarchical update scheme.

Thursday, 26 November · 14:40 – 15:05

Riemannian Multilevel Optimization

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Keywords: Riemannian optimization, multilevel optimization, constrained energy minimization

Multilevel optimization methods are highly effective for discretized energy minimization problems, but their Euclidean formulation does not directly apply to manifold constraints. We introduce a Riemannian extension of multilevel optimization based on a coarse model that is first-order coherent with the fine-level objective and yields descent directions under mild retraction-convexity assumptions. The framework includes metric-compatible vector transfer operators for passing first-order information between levels, covering both intrinsic and extrinsic constructions. We formulate two-level and multilevel algorithms and establish global convergence using a Riemannian Zoutendijk-type argument. Applications to Kohn–Sham density functional theory, Gross–Pitaevskii ground-state computation, and binary continuous cuts demonstrate the method on Stiefel, ellipsoid, and Bernoulli manifolds. Numerical experiments show significant reductions in computational time compared with single-level Riemannian optimization.

This talk is based on joint work with Yara Elshiaty, Jonas Püschel, Tatjana Stykel, and Ferdinand-Joseph Vanmaele; see [1].

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Thursday, 26 November · 15:30 – 16:15

TBA

Michael Hintermüller

Weierstrass Institute (WIAS), Berlin

Thursday, 26 November · 16:15 – 17:00

Locally-Averaged McCormick Relaxations for Discretization-Regularized Inverse Problems

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Keywords: inverse problems, regularization by discretization, McCormick relaxation

As a prototypical coefficient inverse problem, we consider identification of the potential in an elliptic PDE from indirect observations and its regularization by discretization in pre-image space – the so-called least squares version of regularization by projection. We are particularly interested in a situation where the coefficient only takes values in a discrete and finite set. In doing so, we seek to minimize the data misfit over the finite dimensional subspace of piecewise constant function on a given partition of the domain, under the PDE as well as the discrete valuedness as constraints. The resulting algorithmic challenge is twofold:

- the discretization level must be adapted to the noise level to allow us to stably recover coefficient and state in this nonlinear inverse problem;
- the discretization-regularized minimization problem must be solved to global optimality.

For the former, we establish a general result on the least squares method for nonlinear ill-posed operator equations. The latter is not possible by gradient-based nonlinear programming algorithms due to non-convexity. We therefore need a global optimization algorithm like (spatial) branch-&-bound, which requires means to compute (approximate) primal upper and dual lower bounds. In order to obtain dual bounds, we employ a McCormick relaxation, combined with local averaging [2], which reduces the number of degrees of freedom and makes the problem amenable to optimization based bound tightening.

Acknowledgements. This work was supported by the Austrian Science Fund (FWF) under grant 10.55776/F100800.

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Thursday, 26 November · 17:10 – 18:10 · Panel

Point Cloud Registration With Machine Learning

Jürgen Hesser

Heidelberg University

Friday, 27 November · 08:30 – 08:55

Optimal Control of Variational Problems on Manifolds

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Keywords: optimization, optimal control, manifolds

In PDE-constrained optimization a control is applied within a partial differential equation (PDE), e.g., as a force field, to achieve a desired configuration of the PDE solution. The objective is usually to find an optimal balance between control cost and deviation of the controlled state from the desired configuration. We consider this problem class in a setting where the PDE is given by a geometric variational equation and the state is a mapping into a nonlinear manifold. We define a Lagrangian function corresponding to this geometric optimal control problem and derive first-order optimality conditions. Moreover, we present a Lagrange–Newton method for their numerical solution. We introduce an implementable algorithm using a local splitting technique and show its equivalence to Newton's method applied to the derivative of the Lagrangian. The theoretical framework is illustrated by numerical examples, including the optimal control of elastic geodesics and surfaces of minimal energy by force fields and the control of inextensible rods and shells [1].

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Friday, 27 November · 08:55 – 09:40

Algorithmic Differentiation for Bilevel Problems with Non-Unique Lower-Level Solutions

Luca Calatroni

University of Genoa

Many learning approaches to inverse problems are bilevel problems, such as learning regularization parameters or training unrolled networks. Their hypergradients are often computed by algorithmic differentiation: differentiating through the iterations of the algorithm that solves the lower-level problem. This works well when the lower-level solution is unique. In underdetermined or degenerate problems, however, many solutions exist, and the derivative depends on which one the algorithm picks. We show that algorithmic differentiation still makes sense in this case. It converges to the derivative of the solution selected by the algorithm, which reflects the algorithm's geometry and initialization. We make this precise for (preconditioned) gradient descent, discuss extensions to mirror descent for constrained problems, and illustrate the results on exemplar inverse problems.

This is joint work with C. Daniele and S. Villa (MaLGa, University of Genoa), and S. Vaiter (CNRS).

Friday, 27 November · 09:40 – 10:05

Locally-Supervised Global Image Reconstruction

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Keywords: inverse problems, partially-supervised learning, invariances, image upsampling, photoacoustic microscopy

We consider the recovery of an image $x \in \mathbb{R}^I$ from incomplete observations $y = P_\Omega x$ with observed pixels $\Omega \subseteq I$, which covers upsampling as well as inpainting. Learning-based solutions approximate the minimizer $\mathbb{E}[X \mid X_\Omega]$ of the supervised loss $\mathbb{E}[\|f(X_\Omega) - X\|^2]$, requiring fully sampled ground truth; self-supervised methods avoid this, but rely on Ω being random and covering the whole index set in expectation. We instead study fixed patterns with incomplete coverage, compensating for the missing randomness by invariances of the image distribution. Our main contributions are the following:

- We introduce a locally-supervised framework based on a fixed subsampling set Ω, a small supervision set $B \subseteq I$, and transformations $(T_\ell)_\ell$ leaving X and $M_\Omega \odot X$ invariant in distribution, with $(T_\ell(B))_\ell$ partitioning I.
- We establish that the optimal reconstruction function is $\mathcal{T}$-equivariant and minimizes the locally supervised loss,

$$\mathbb{E}[X \mid X_\Omega] = \operatorname*{arg\,min}_{f \in \mathcal{F}(\mathbb{R}^I;\mathcal{T})} \mathbb{E}\big[\|M_B \odot f(X_\Omega) - X_B\|^2\big],$$

  so supervision on the small subset B recovers the fully supervised solution.
- We derive a sparse-dense sampling scheme [1] with supervision on a single quadrant, realized by a U-Net with circular padding.
- Experiments on optical-resolution photoacoustic microscopy of lung tissue show performance close to fully supervised training at a fraction of the measurements, outperforming patch-wise training.

Future work includes noise, invariances beyond translations, and extensions to ill-posed inverse problems. This is joint work with B. Walder, D. Toader, R. Nuster, G. Paltauf, P. Burgholzer, G. Langer and L. Krainer.

Acknowledgements. Supported by the Austrian Research Funding Association (FFG), project Compressed Photoacoustic Remote Sensing (FO999898886).

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Friday, 27 November · 10:30 – 10:55

Atomic Gradient Flows: Gradient Flows on Sparse Representations

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Keywords: optimization, inverse problems, gradient flows in metric spaces, sparsity

One of the most popular approaches for optimization problems in the space of measures are the so-called Particle Gradient Flows (PGFs). PGFs restrict the problem to linear combinations of Dirac deltas and then perform a gradient flow in the space of weights and positions [2]. In this talk, I will show how to generalize PGFs to arbitrary spaces and optimization problems:

$$\inf_{u\in X} G(u) \quad \text{where} \quad G(u) := F(Au) + R(u)$$

where X is a Banach space, F is a smooth fidelity term, A is a linear operator and R is a convex regularizer [1]. The crucial idea is how to choose the right notion of particles, which we argue to be the extremal points of the ball of R. Such observation is motivated by Krein–Milman theorem that allows to represent any element in the domain of G as the limit of convex combinations of such extremal points. In particular, one considers the restriction of G to *sparse representations*

$$u_{\text{sparse}} = \sum_{i=1}^{n} c_i u_i \quad c_i \in \mathbb{R}_+, \quad u_i \in \text{Ext}(\{R \leq 1\})$$

and performs a gradient flow of such restriction in the extremal points u_i and coefficients c_i, named *Atomic Gradient Flow* (AGF). I will consider analytical properties of such flow, showing its well-posedness as a minimizing movement scheme, its uniqueness and the related contraction estimates. Then, using Choquet's theorem, I will present how to lift the problem into the Wasserstein space over extremal points and weights, discussing the relations between the lifting of the AGF and the AGF itself. Finally, I will show how AGFs can be used to deal with optimization problems with BV-seminorm and Optimal Transport regularizations.

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Friday, 27 November · 10:55 – 11:40

Challenges in Dynamic Inverse Problems

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Keywords: inverse problems with inexact forward operator, sequential subspace optimization, motion estimation and compensation, dynamic tomography

Dynamic inverse problems arise whenever the searched-for quantity changes during the data acquisition. They play an important role in medicine and biology, for example in imaging physiological processes or compensating for patient motion, as well as in non-destructive testing, where objects may be imaged under changing mechanical loads.

Due to the dynamic behavior of the object, the resulting measurements are no longer consistent with a single static model. Ignoring this inconsistency and applying standard regularization methods typically leads to a difficult trade-off between spatial and temporal resolution. More importantly, temporal changes can manifest themselves as severe motion artefacts, potentially obscuring relevant structures and compromising further image analysis steps.

In this talk, we give an overview of two complementary ways to incorporate dynamics into the regularization process. The first treats the unknown dynamics as inexactness of the forward model and is accounted for via regularized sequential subspace optimization. The second approach exploits additional information about the underlying motion, therefore requiring explicit motion estimation and compensation strategies. Both approaches will be evaluated on simulated and real data from different imaging applications.

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Friday, 27 November · 13:10 – 13:35

Derivative-Free Bilevel Learning for Inverse Problems: From Stochastic Gradient Estimation to Multilevel Methods

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Keywords: bilevel optimization, derivative-free optimization, multilevel Monte Carlo, Gaussian smoothing, inverse problems

We consider bilevel learning problems arising in inverse problems, where regularization parameters are optimized from data and each upper-level function evaluation requires the approximate solution of a lower-level reconstruction problem. When derivatives of the lower-level solution map are unavailable or undesirable, zeroth-order methods provide a flexible alternative, but can become computationally expensive due to increasingly accurate inner solves. We propose a multilevel Monte Carlo approach for derivative-free bilevel optimization that combines Gaussian smoothing with a hierarchy of lower-level accuracies. The resulting estimator uses many inexpensive coarse solves and only few accurate fine solves. We derive convergence and complexity guarantees and identify the properties of the lower-level solver that determine whether multilevel sampling yields a genuine computational speedup. Numerical experiments for imaging inverse problems illustrate the potential gains.

Friday, 27 November · 13:35 – 14:20

From Inverse Problems in PDEs to Convex Semidefinite Optimization

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Keywords: inverse problems, elliptic PDEs, semidefinite optimization

Several applications in medical imaging and non-destructive material testing lead to inverse elliptic coefficient problems, where an unknown coefficient function in an elliptic PDE is to be determined from partial knowledge of its solutions. This is usually a highly non-linear ill posed inverse problem, for which unique reconstructability results, stability and resolution estimates and global convergence of numerical methods are very hard to achieve.

After FEM discretization [1], inverse elliptic coefficient problem can often be written in the following form:

$$\text{Reconstruct} \quad \hat{y} \in \mathbb{R}^m_+ \quad \text{from} \quad \hat{Y} := F(\hat{y}) \in \mathbb{S}^n,$$

where $F: \mathbb{R}^m_+ \to \mathbb{S}^n_+$ is given by

$$F(y) = -BA_y^{-1}B^T, \quad A_y := \sum_{j=1}^m y_j A_j,$$

with $A_j \in \mathbb{S}^d_-$, and surjective $B \in \mathbb{R}^{n\times d}$ $(d, n, m \in \mathbb{N})$.

The matrix $-F(y)$ is the Schur complement of the block matrix $\left(\begin{smallmatrix} -A_y & B^T \\ B & 0 \end{smallmatrix}\right)$, which yields that F is convex and monotonically non-increasing with respect to the semi-definite ordering of symmetric matrices (aka Loewner order). A further Schur complement argument shows that $F(y) \preceq Y$ is equivalent to $A_y \preceq C := -B^T Y^{-1} B$, cf. [4]. Hence, the set of all $y \in \mathbb{R}^m_+$ satisfying $F(y) \preceq Y$ is convex. The solution of the inverse problem $F(y) = Y$ lies on a corner of this convex set with maximally active constraints.

In this talk, we will build upon this relation to discuss reformulations of the nonlinear inverse problem $F(y) = Y$ as a semidefinite optimization problem [2, 3]

$$b^T y \to \max! \quad \text{s.t.} \quad \sum_{i=1}^m y_i A_i \preceq C.$$

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Friday, 27 November · 14:20 – 15:05

Quantitative Convergence of Proximal Splitting Iterations in Uniformly Convex Metric Spaces

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Keywords: optimization, metric space, proximal mapping, operator splitting

We provide sufficient conditions for quantitative convergence of the iterates of proximal splitting algorithms for minimizing a sum of functions on a metric space.

The setting for our development is a p-uniformly convex metric space, denoted (G, d). Our central goal is to solve the following optimization problem

$$\operatorname*{argmin}_{x\in G} \sum_{j=1}^{m} f_j(x).$$

For geodesic metric spaces, in the absence of linear structure *implicit tools* using the *proximal mapping* [10] are well-developed and provide notions of descent based on *gradient flows* [8, 1]. This is applied to evolution systems (see for instance [9]) and metric measure spaces (see for instance [13]). Purely implicit methods based on proximal splitting are possible whenever the proximal mapping can be computed explicitly [3, 11, 5, 12].

Let

$$\begin{aligned}\overline{T} \coloneqq (\tau_m \operatorname{prox}^p_{f_m,\lambda_m} \oplus(1-\tau_m)\operatorname{Id}) \circ \cdots \\ \circ (\tau_2 \operatorname{prox}^p_{f_2,\lambda_2} \oplus(1-\tau_2)\operatorname{Id}) \circ (\tau_1 \operatorname{prox}^p_{f_1,\lambda_1} \oplus(1-\tau_1)\operatorname{Id}).\end{aligned}$$

Holding the parameters τ_j and λ_j fixed, we characterize the properties of the mapping $\overline{T}$ and determine the convergence properties of fixed point sequences (x^k) under $\overline{T}$:

$$x^{k+1} = \overline{T}x^k.$$

The theory does not assume that the functions f_j have common minima, nor does it require vanishing proximal parameters or step sizes. Our results are stated for general p-uniformly convex spaces with curvature bounded above, and a corollary specializes the main theorem to Hadamard spaces, where many assumptions for the more general setting can be dropped. The theory is demonstrated with computation of Fréchet means in the space of SPD matrices with the affine invariant metric (a Hadamard space) and the sphere with the usual geodesic metric (a CAT(κ) metric space).

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